

A Novel Robust Visible Light Positioning Based on Sparsity-Adaptive Compressed Sensing

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Abstract—The fingerprint-based visible light positioning (VLP) method offers high-precision indoor positioning with centimeter-level accuracy, making it well-suited for applications like robot navigation. However, the sensitivity of photodiodes (PDs) to natural light compromises the system's robustness. To address this challenge, we construct a robust received signal strength (RSS) database that leverages the properties of pulse width modulation (PWM) technology to minimize interference from natural light. This robust RSS database serves as the foundation for a novel, low-complexity, sparsity-adaptive compressed sensing (CS) algorithm, which enables accurate signal reconstruction without requiring prior knowledge of sparsity. Experimental results show that the proposed VLP system effectively mitigates the impact of natural light and maintains consistent positioning performance across varying natural lighting conditions. With a grid spacing of 20 cm, the proposed RSS database construction method achieves an average positioning error (APE) of 1.36 cm in VLP, marking a 75.54% reduction compared to the traditional RSS database construction method. Furthermore, the proposed VLP system maintains an APE of 2.15 cm even at a 40 cm grid spacing. This demonstrates that our system achieves excellent positioning accuracy while maintaining low computational and storage complexity.

Index Terms—Compressed sensing (CS), fingerprint localization, natural light, received signal strength (RSS), visible light positioning (VLP).

I. INTRODUCTION

THE development of the Internet of Things (IoT) has increased the demand for high-precision positioning systems

in indoor applications such as robot navigation and automatic parking [1]. Leveraging the advantages of light-emitting diode (LED) – including low power consumption, long service life, and lack of electromagnetic interference – visible light positioning (VLP) systems based on LED lighting have emerged as a promising solution for precise indoor positioning [2], [3]. In VLP systems, LED light signals serve as transmission sources, which mobile terminals, such as robots, can detect using integrated photodiodes (PDs) or cameras. While a camera-based VLP offers higher positioning accuracy than PD-based systems, it consumes more power and raises potential privacy concerns [4], [5].

Received signal strength (RSS)-based VLP has gained significant attention for its simplicity and cost-effectiveness in positioning. This approach is generally categorized into two methods: ranging and fingerprinting [6], [7]. Unlike the ranging method, which requires precise signal propagation modeling, the fingerprinting method is more practical and widely used for indoor location estimation [7]. Although increasing the density of a fingerprint database improves positioning accuracy, it also demands considerable time and labor to build a comprehensive RSS database. To alleviate this burden, recent efforts have focused on regenerating fingerprint data in a fingerprint-based VLP system [3], [7], [8], [9], [10], [11].

Thus, developing a reliable RSS database remains a critical technical challenge for fingerprint-based VLP systems. Although interference from multiple LED emission sources can be minimized using various visible light communication (VLC) protocols such as time-division multiplexing (TDM) and frequency division multiplexing (FDM) [8], [10], [11], [12], [13], [14], [15], these methods are ineffective against the influence of natural light – particularly in dim indoor environments where ambient daylight still plays a role. Natural light fluctuates both spatially and temporally, and discrepancies between lighting conditions during offline training and online positioning can cause significant positioning errors (PEs). Therefore, a robust VLP system must account for the dynamic nature of natural light to maintain positioning accuracy.

The noise measurement and mitigation scheme proposed in [16] utilizes Allan variance; however, the experiments were conducted in a dark room. To address the influence of ambient light, a fusion of a PD and a camera for VLP (FusionVLP) was proposed in [17]. While using camera technology compensates for the PD's sensitivity to environmental changes, this integration increases both equipment costs and power consumption. In [18], the authors minimize noise impact by leveraging

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differences in received voltage levels, but this approach requires LED transmitters to support multi-level modulation. Furthermore, [19] provides a detailed analysis of how ambient light affects the traditional Lambert model and proposes a distance ratio-based VLP (RatioVLP) method to circumvent these issues in conventional distance estimation. However, this method depends on the accurate calibration of parameters. Although Kalman filtering can improve the accuracy of RSS measurements in VLP systems, it remains vulnerable to interference from natural light [20].

Our previous work [12] developed a practical VLP experimental platform, but the system design did not account for the effects of natural light. While some experimental platforms demonstrate high positioning accuracy, they overlook the influence of fluctuating natural light on VLP performance [11], [12], [15], [16], [17], [20], [21], [22], [23], [24], [25]. One common approach to detect the light intensity of LED emission sources is to measure the background light first by turning off all LEDs and then sequentially activating each LED for individual measurements [10], [22]. However, this method requires additional time to detect background light and lacks unique coding for each LED, resulting in limited interference resistance. Moreover, an oscilloscope is required to measure light intensity, and the impact of varying natural light remains unexamined.

This paper focuses on mitigating the influence of natural light in fingerprint-based VLP systems, considering two key challenges: (i) Natural light fluctuates over time, affecting most indoor spaces that are not completely dark. (ii) Natural light varies spatially, creating uneven distributions across different areas, even simultaneously. The most common method for regulating LED brightness in VLP systems is pulse-width modulation (PWM) [26], widely used in prior works [11], [12], [16], [20], [21]. This paper introduces a robust method for constructing an RSS-based database by leveraging PWM characteristics. The proposed approach effectively mitigates the impact of natural light without requiring any hardware modifications.

VLP based on fingerprint is a data-driven approach, making machine learning (ML) well-suited for this task. Typical methods include random forest (RF), extreme learning machine (ELM), support vector machine (SVM), multi-layer perceptron (MLP), weighted K -nearest neighbor (WKNN), and extreme gradient boosting (XGBoost) [3], [27], [28]. Classifier-based methods such as RF, ELM, and SVM rely on ultra-dense RSS databases. In contrast, WKNN can estimate positions at non-grid points, achieving higher accuracy even with reduced grid density. As a result, WKNN is widely adopted in fingerprint-based VLP systems [7], [8], [11], [13], [29], [30]. However, environmental factors such as natural light can degrade positioning performance. To address this, studies [29] and [30] proposed fingerprint database methods based on a multi-pixel photon counter (MPPC) and center pixel coordinate (CPC), demonstrating their effectiveness via WKNN-based evaluation. MLP, as a classic ML algorithm, has been widely used for VLP system performance evaluation [3], [11], [13], [31]. However, its performance is highly dependent on training data volume. To address environmental challenges, the authors of [32] developed a robust VLP system named DeepVLP. However, its performance

depends on training data that covers diverse lighting conditions to achieve robustness.

Fingerprint-based localization requires discretizing the space into a grid structure, with only a small subset of grid points contributing to the target's localization. This subset is significantly smaller than the number of grid points, satisfying the sparsity condition required by compressed sensing (CS) [33]. An indoor reverse VLP system using the reweighted ℓ_1 -norm minimization (RW ℓ_1 M) method was proposed in [34]. To address the non-convexity challenge in RSS-based VLP, [35] introduced a grid sampling scheme, dividing the space into smaller grid cells. As a solution, the orthogonal matching pursuit (OMP) and proximity-based methods can separate LED signals and estimate receiver positions [36]. Building on [36], the authors of [37] proposed a novel VLP framework based on CS. Additionally, a CS-based VLP system has been proposed for accurate simultaneous localization of multiple robots [38], [39], but it requires prior knowledge of sparsity, i.e., the number of co-located robots. As the number of robots increases, a denser LED deployment becomes necessary. Although CS has been applied in VLP systems [34], [35], [36], [37], [38], [39], it still faces several challenges: (i) They are limited to specific indoor scenarios, requiring a large number of LEDs or PDs. (ii) Using fixed sparsity or reliance on prior sparsity information compromises the system's robustness. (iii) The reported results are based solely on simulations, with no experimental validation provided.

Building on the abovementioned challenges, this paper develops a robust RSS database for VLP and introduces a novel, low-complexity, sparsity-adaptive CS technique. The main contributions of this work, in comparison with existing VLP methods, are summarized as follows.

- *Robust RSS Database Construction:* A robust RSS database for VLP is built using PWM, the most widely adopted technology for LED brightness modulation. The real-time residual signal between high-level and low-level sampling signals is used to minimize interference from natural light. The proposed method requires no additional hardware or complex filtering algorithms, effectively mitigating the impact of natural light by capitalizing on the inherent characteristics of PWM. This solution achieves both cost efficiency and low implementation complexity.
- *Low-complexity and Sparsity-Adaptive CS Algorithm:* A novel, low-complexity, sparsity-adaptive CS algorithm is introduced in the VLP, capable of accurately reconstructing signals without prior knowledge of their sparsity. By exploiting the sparsity of location fingerprints, the localization problem is reformulated as a sparse matrix reconstruction task. The algorithm adaptively calculates the sparsity based on the residual value of the reconstructed signal, maintaining low computational complexity.
- *Experimental Validation:* The experimental results demonstrate that the proposed RSS database construction method effectively mitigates natural light interference. The method consistently delivers high positioning accuracy even under significant variations in natural light intensity between offline training and online positioning phases. Furthermore, the proposed VLP system exhibits exceptional robustness

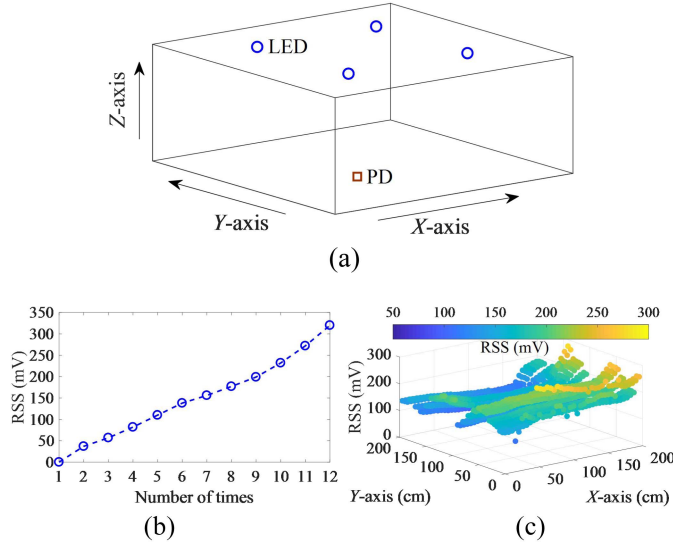


Fig. 1. The distribution of natural light intensity over time and space: (a) Typical PD-based indoor VLP experimental system model (b) changes over time (c) changes over space.

and achieves efficient localization with minimal computational overhead.

The rest of this paper is organized as follows: Section II introduces the proposed robust VLP based on low-complexity and sparsity-adaptive CS. Section III presents the experimental results. Finally, Section IV concludes the paper.

II. DESIGN OF A ROBUST VLP USING LOW-COMPLEXITY AND SPARSITY-ADAPTIVE CS

This section examines the impact of natural light on visible light RSS and introduces a robust RSS database for VLP to mitigate its effects. Next, the fingerprint-based VLP problem is formulated, highlighting its inherent challenges. To address these challenges, a novel low-complexity, sparsity-adaptive CS algorithm is proposed.

A. Analysis of Natural Light Intensity Variations Over Time and Space

A typical PD-based indoor VLP experimental system model is shown in Fig. 1(a). The PD model used for measurement the intensity of natural light is OPT101, with movement controlled by a computer numerical control (CNC) machine. It is worth noting that in order to analyze the change of natural light over time and space, all LEDs are turned off. Fig. 1(b) illustrates the natural light intensity at 12 different time points collected at the same location, revealing a range from low to high values. As shown in Fig. 1(b), the light intensity measures 0 mV during nighttime or under low-light conditions, but it can reach up to 320 mV during daylight hours. The two-dimensional (2D) distribution of natural light intensity over a $2\text{ m} \times 2\text{ m}$ area, collected at similar times, is depicted in Fig. 1(c). This figure highlights the non-uniform distribution of light intensity across different spaces. Hence, considering the impact of natural light is crucial when building an RSS database. If the lighting conditions

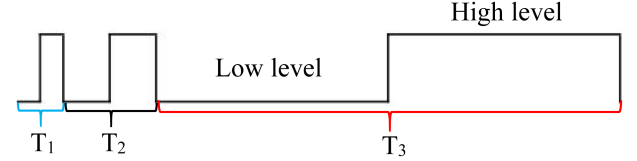


Fig. 2. LED positioning signal encoding format.

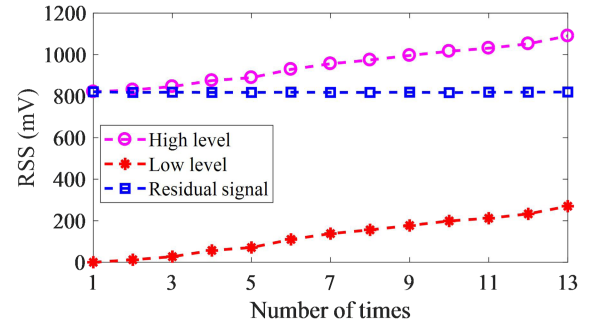


Fig. 3. RSS variation over sampling time.

during online positioning differ from those used during offline training, significant PEs may occur.

B. A Low-Complexity RSS Database Construction for Mitigating Natural Light Interference

The proposed method for constructing a robust RSS database based on RSS differences utilizes the characteristics of PWM technology to mitigate the impact of natural light effectively. While PWM has been widely used in VLP systems (e.g., [11], [12], [16], [20], [21]), prior studies have not addressed natural light interference through PWM technology. Fig. 2 illustrates the encoding format for the LED positioning signal, in which each LED transmits positioning signals across three timeslots. In T_1 , a PWM signal with a frequency of 10 kHz and a 50% duty cycle represents the starting bit of the positioning signal. T_2 is used to transmit LED identification (ID) codes for four LEDs assigned frequencies of 2 kHz, 3 kHz, 4 kHz, and 5 kHz, respectively. T_3 uses 1 kHz PWM with a duty cycle of 50% to measure low and high voltage values, obtaining the corresponding RSS values after analog-to-digital conversion. The real-time residual signal between high-level and low-level sampling signals is used to mitigate natural light interference effectively: low-level signals represent natural light intensity, while high-level signals reflect the combined LED signals and natural light intensity. As shown in Fig. 3, both high and low-level signals increase with rising natural light intensity under constant transmission power and test point location. However, the residual signal between them remains almost constant, making it possible to mitigate the impact of natural light by using the difference between high-level and low-level signals.

C. Problem Formulation

The 2D space is assumed to be uniformly divided into a grid, with each grid point designated as a training fingerprint. Assume that there is a total of N training fingerprints and the

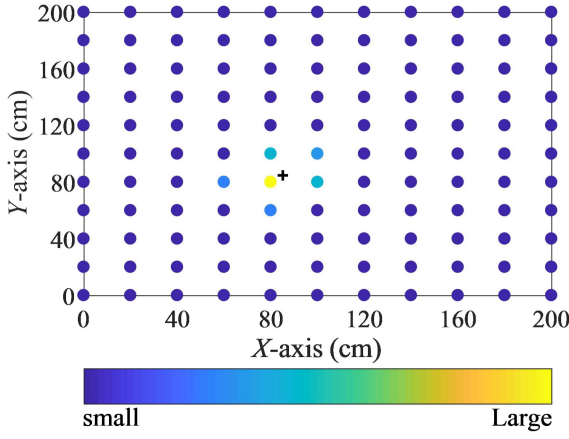


Fig. 4. An example of a sparse structure in a fingerprint-based VLP model. The plus sign marks the target's position, while the circle indicates the position of a training fingerprint. The degree of correlation between the training fingerprint and the target, referred to as the energy value, is represented through varying colors.

weight vector of the fingerprints, i.e., the unknown signal, is $\theta \in \mathbb{R}^{N \times 1}$. Applying CS principles concludes that the fingerprints with higher energy are more strongly correlated with the target. Consequently, only the K training fingerprints with the highest correlation need to be selected for accurate localization, a concept referred to as K -sparsity. Fig. 4 illustrates an example of a sparse structure within a fingerprint-based VLP model. When the training fingerprint with the highest energy value is selected as the target coordinate, it is termed 1-sparsity. However, since the target may appear randomly within the positioning area, the vector θ has K peaks – referred to as K -sparse. Because of the target's random movement within the positioning space, managing sparsity is challenging in CS-based localization methods. Accurate signal reconstruction requires careful selection of sparsity: if it's too low, reconstruction accuracy suffers, whereas if it's too high, weakly correlated fingerprints may be introduced, reducing the accuracy and robustness of the CS algorithm. Therefore, a sparsity-adaptive approach that does not require prior knowledge of sparsity is essential to improve the robustness of VLP.

Let $\mathbf{x}_i = (x_{1i}, x_{2i}, x_{3i})^T$, for $i = 1, 2, \dots, N$, denote the 3-D coordinates of the i th training fingerprint, where the superscript 'T' indicates the transpose. The matrix $\mathbf{X}$, representing the coordinates of N training fingerprints, is given by

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N] = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1N} \\ x_{21} & x_{22} & \dots & x_{2N} \\ x_{31} & x_{32} & \dots & x_{3N} \end{bmatrix}. \quad (1)$$

The RSS matrix $\mathbf{A}$ regarding N training fingerprints is defined as

$$\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_N] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1N} \\ a_{21} & a_{22} & \dots & a_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ a_{M1} & a_{M2} & \dots & a_{MN} \end{bmatrix}, \quad (2)$$

where $\mathbf{a}_i = (a_{1i}, a_{2i}, \dots, a_{Mi})^T$ represents the RSS vector of the i th training fingerprint and a_{mi} denotes the RSS residual

signal received by the PD from the m th LED at the i th training fingerprint, $m = 1, 2, \dots, M$. In this paper, $a_{mi} = a_{mi}^H - a_{mi}^L$, where a_{mi}^H and a_{mi}^L represent the RSS vectors of the high-level and low-level signals, respectively, as illustrated in Fig. 2.

Assuming the 3-D coordinates of the target are represented by $\mathbf{y} = (y_1, y_2, y_3)^T$, we get

$$\mathbf{y} = \mathbf{X}\theta. \quad (3)$$

where $\theta = (\theta_1, \theta_2, \dots, \theta_N)^T$ represents the weight vector of N training fingerprints. Because the target coordinates are unknown, the value of the weight vector θ cannot be determined using (3).

In the online phase, we assume that the user equipment (UE) receives the RSS residual signal vector $\mathbf{b} = (b_1, b_2, \dots, b_M)^T$ from M LEDs, where $b_m = b_m^H - b_m^L$, with b_m^H and b_m^L representing the RSS vectors of the high-level and low-level signals, respectively. The positioning criterion in the fingerprint method is to determine the weight vector θ of the N training fingerprints based on the RSS vector $\mathbf{b}$ and the RSS matrix $\mathbf{A}$ of the N training fingerprints, given that

$$\mathbf{b} = \mathbf{A}\theta, \quad (4)$$

In most cases, indoor LED distribution is sparse, whereas the density of training fingerprints is relatively high. Consequently, the problem in (4) becomes challenging as the number of measurements M is significantly smaller than the number of unknown variables N . However, with properly configured LED quantity and layout, the resulting RSS vectors demonstrate spatial uniqueness across training positions. This means only a few training fingerprints exhibit a strong correlation with the target (i.e., the signal θ is K -sparse). Thus, the fingerprint weight vector estimation can be reformulated as a sparse vector reconstruction problem. CS capitalizes on this inherent sparsity characteristic, requiring only a small number of measurements to preserve the essential information of the original fingerprint weight vector. Thus, the unknown signal θ can be reconstructed using the least square estimation method [40], expressed as

$$\theta = \arg \min_{\theta} \|\mathbf{b} - \mathbf{A}\theta\|_2^2, \quad (5)$$

where $\|\cdot\|_2$ represents the ℓ_2 -norm operation of a vector.

To scale the correlation between the training fingerprint and the target within a range of 0 to 1, a ℓ_2 -norm normalization operation is applied on both vectors $\mathbf{a}_i$ and $\mathbf{b}$, given by

$$\alpha_i = \left[\frac{a_{1i}}{\|\mathbf{a}_i\|_2}, \frac{a_{2i}}{\|\mathbf{a}_i\|_2}, \dots, \frac{a_{Mi}}{\|\mathbf{a}_i\|_2} \right]^T, \quad (6)$$

$$\beta = \left[\frac{b_1}{\|\mathbf{b}\|_2}, \frac{b_2}{\|\mathbf{b}\|_2}, \dots, \frac{b_M}{\|\mathbf{b}\|_2} \right]^T. \quad (7)$$

This yields a new RSS matrix $\mathbf{C} = [\alpha_1, \alpha_2, \dots, \alpha_N]$, allowing (5) to be redefined as

$$\theta = \arg \min_{\theta} \|\beta - \mathbf{C}\theta\|_2^2. \quad (8)$$

D. Analysis of the Correlation Degree Between Vectors α_i and β

The correlation degree between vectors α_i and β is defined as

$$u_i = \alpha_i^T \beta = |\alpha_i| |\beta| \cos(\alpha_i, \beta). \quad (9)$$

where $|\cdot|$ denotes the modulus operation of a vector, equivalent to calculating the vector's ℓ_2 -norm and (α_i, β) represents the angle between vectors α_i and β .

Given that each element in the vectors α_i and β represents a voltage value greater than 0, $u_i \in (0, 1]$ can be determined. A larger u_i indicates a stronger correlation between vectors α_i and β , signifying a greater contribution of vector α_i to the weight vector θ .

Challenge: From (9), it is clear that obtaining the K training fingerprints with the highest correlation to the vector β is sufficient to reconstruct the K -sparse signal θ . However, due to the random movement of the UE, the measurement vector β changes with spatial variation, making it difficult to determine an optimal K value.

E. A Low-Complexity and Sparsity-Adaptive CS

To address the above challenge, this work introduces a novel, low-complexity, sparsity-adaptive CS algorithm that enables precise signal reconstruction without requiring prior knowledge of signal sparsity, thereby greatly enhancing the robustness of the VLP system. The pseudocode for the proposed sparsity-adaptive CS method is detailed in Algorithm 1.

In Algorithm 1, line 2 identifies the $K_{\max}$ largest values in vector $\mathbf{u}$, with $\mathbf{I}$ representing the corresponding index set. Line 3 uses this index set $\mathbf{I}$ to select the $K_{\max}$ training fingerprints that have the highest correlation between vectors α_i and β , forming a new matrix Ψ . Lines 4 to 13 outline the process for determining the optimal sparsity K based on the matching residual e . Line 8 highlights that the proposed algorithm operates as a greedy selection strategy. As the value of k increases, the matching error e progressively decreases until convergence is achieved. Then, the optimal sparsity can be obtained based on the matched error e , thereby enhancing the robustness of the algorithm. Finally, lines 14 to 16 calculate the weight vector $\theta_{K \times 1}$ for the K optimal training fingerprints according to the determined sparsity, yielding coordinate $\mathbf{y}$ of the target.

Compared with OMP [40], the key innovations of the proposed Algorithm 1 are twofold: (i) Each OMP iteration involves computing the inner products between all atoms in the dictionary and the current residual. Once incorrect atoms are included in the support set, the residual update is computed based on an erroneous subspace, leading to error propagation. The proposed Algorithm 1 determines the candidate support set through a single global correlation screening, eliminating the need for repeated inner product calculations, reducing the complexity from $KO(MN)$ (for K iterations) to a single $O(MN)$ operation; (ii) The proposed Algorithm 1 introduces an adaptive sparsity selection mechanism, which dynamically adjusts the sparsity level based on the matching error. This eliminates the need for manual threshold tuning, ensures optimal approximation

Algorithm 1: The Pseudocode of the Low-Complexity and Sparsity-Adaptive CS.

Input: RSS matrix $\mathbf{C}$, RSS vector β , the maximum sparsity $K_{\max}$.

Output: the coordinate $\mathbf{y}$ of the target.

Initialize: Index set $\mathbf{I} = \emptyset, \theta = \mathbf{0}_{N \times 1}$.

1. $\mathbf{u} = (u_1, u_2, \dots, u_N)^T = \mathbf{C}^T \beta$;
 2. $\mathbf{I} = \text{Findmax}(\mathbf{u}, K_{\max})$;
 3. $\Psi(:, 1 : K_{\max}) = \mathbf{C}(:, \mathbf{I}(1 : K_{\max}))$;
 4. **for** $k = 1 : K_{\max}$
 5. $\theta_{k \times 1} = \arg \min_{\theta_{k \times 1}} \|\beta - \Psi(:, 1 : k) \theta_{k \times 1}\|_2^2$;
 6. $\mathbf{R} = \beta - \Psi(:, 1 : k) \theta_{k \times 1}$;
 7. $e(k) = \mathbf{R}^T \mathbf{R}$;
 8. **if** $(k > 1) \& \& [e(k) > e(k-1)]$
 9. $K = k - 1$;
 10. **break**
 11. **end if**
 12. $K = k$;
 13. **end for**
 14. $\theta_{K \times 1} = \arg \min_{\theta_{K \times 1}} \|\beta - \Psi(:, 1 : K) \theta_{K \times 1}\|_2^2$;
 15. $\theta(\mathbf{I}(1 : K)) = \theta_{K \times 1}$;
 16. **Output:** $\mathbf{y} = \mathbf{X} \theta$;
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for each test point, and enhances the algorithm's robustness in varying scenarios.

F. Computational Complexity Analysis of the Proposed Algorithm

In Algorithm 1, the time complexity for calculating $\mathbf{C}^T \beta$ is $O(MN)$. In line 2, finding the $K_{\max}$ largest values in the vector $\mathbf{u}$ has the time complexity of $O(K_{\max}N)$. The time complexity for lines 3 to 15 can be disregarded as $K \leq K_{\max} \ll N$ and $M \ll N$ in practical applications. In line 16, the time complexity is $O(3N)$. Thus, the overall computational complexity of the proposed Algorithm 1 is approximately $O(MN + K_{\max}N + 3N)$. The main difference between the computational complexity of the proposed Algorithm 1 and OMP algorithm [40] is that each OMP iteration involves computing the inner products between all atoms in the dictionary and the current residual, with a per-iteration time complexity of $O(MN)$, and the OMP algorithm using a fixed threshold needs to perform K iterations, so the computational complexity of OMP algorithm is about $O(KMN + KN + 3N)$. In summary, both the proposed and OMP algorithms exhibit comparable computational complexity, primarily determined by the number of training fingerprints N in the actual scene.

III. EXPERIMENTAL RESULTS

Experiments were conducted to evaluate the performance of the proposed sparsity-adaptive CS method. The setup, illustrated in Fig. 5, includes four LEDs, a CNC machine, four LED driver boards, and a receiver module. The test area is a $2 \text{ m} \times 2 \text{ m}$ 2D space, with a vertical distance of 1.8 m between the four LEDs

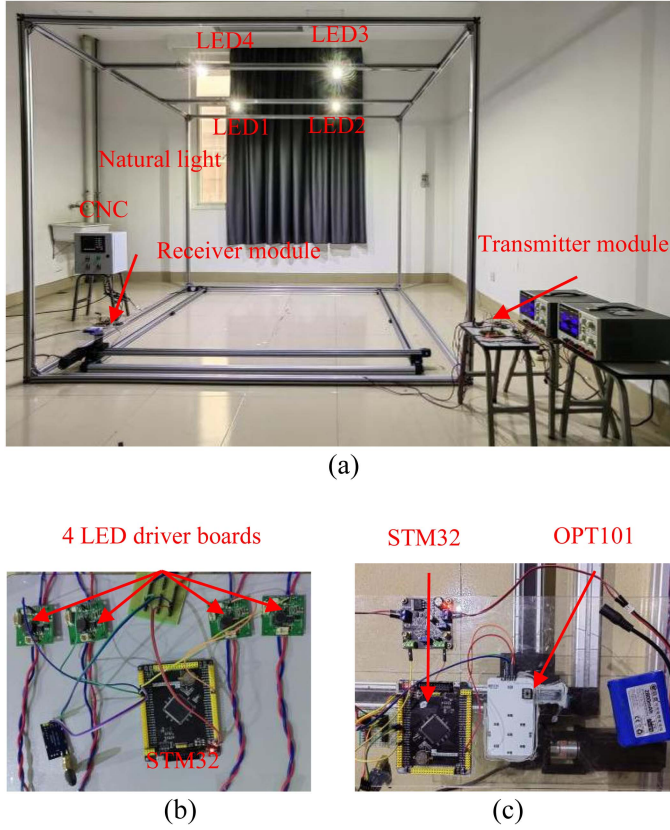


Fig. 5. Experimental setup. (a) VLP experimental platform. (b) Transmitter module. (c) Receiver module.

and one PD. The LED coordinates are positioned at (0.5 m, 0.5 m), (0.5 m, 1.5 m), (1.5 m, 1.5 m), and (1.5 m, 0.5 m). Each LED is a 10 W CREE XML2, and the receiver module comprises an STM32 and an OPT101 PD. As shown in Fig. 5(c), the receiver is mounted on the CNC machine to maintain a constant orientation for the OPT101, pointing directly upward. This configuration is applicable to mobile robot positioning [3], [16], [17], [18], [19], [25].

The OPT101 processes received signals via two channels: one for analog-to-digital conversion to generate the required RSS values for positioning and the other for LED ID code identification. All RSS values are stored on an SD card. Fig. 6 illustrates the distribution of LEDs, training fingerprints, and test points with a grid spacing S of 20 cm. A total of 1600 test points were evenly distributed within the 2D space. The CNC machine measured the actual positions of training fingerprints and test points with a 0.1 mm accuracy, enabling automatic and precise coordinate logging and PE assessment of the test point. Positioning signals were transmitted through time-division multiplexing by the four LEDs, as shown in Fig. 7, while Fig. 2 illustrates the encoding format of the LED positioning signal. In the proposed low-complexity and sparsity-adaptive CS approach, the maximum sparsity $K_{\max}$ is set to 7.

Fig. 8 shows the contour map of natural light intensity measured in the experiment. Fig. 8(a) and (b) show the distribution of natural light intensity at the test points under two conditions, labeled Scenario 1 and Scenario 2, respectively. Fig. 8(c) displays

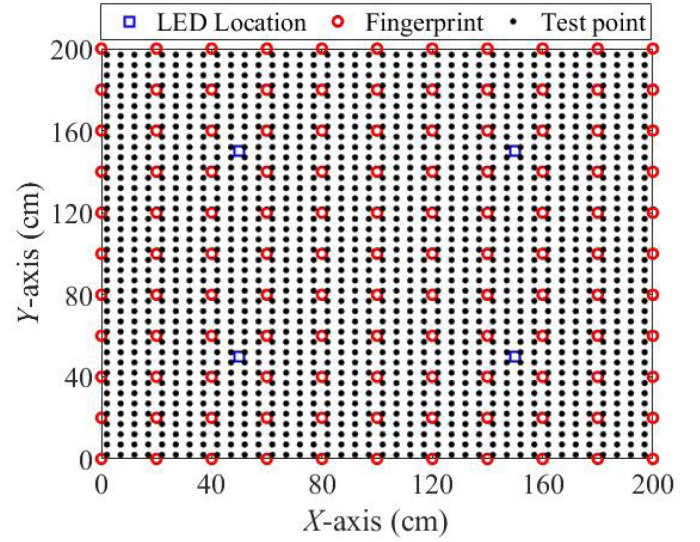


Fig. 6. Layout of LED positions, training fingerprints, and test points.

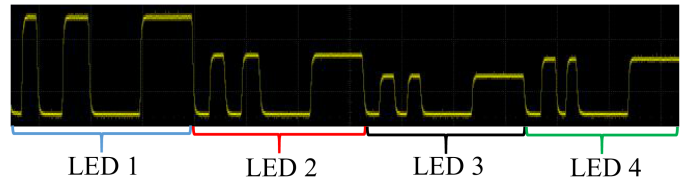


Fig. 7. Time-division multiplexing scheme for Four LEDs.

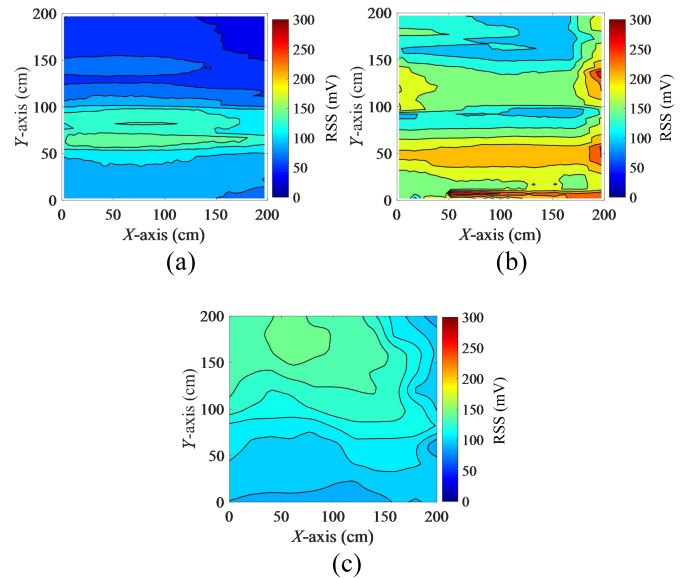


Fig. 8. The contour map of the intensity of natural light used in the experiment. (a) Scenario 1. (b) Scenario 2. (c) Training fingerprints.

the natural light intensity distribution of the training fingerprints. As is shown, the natural light intensity distribution of the test points in both scenarios differs significantly from that of the training fingerprints.

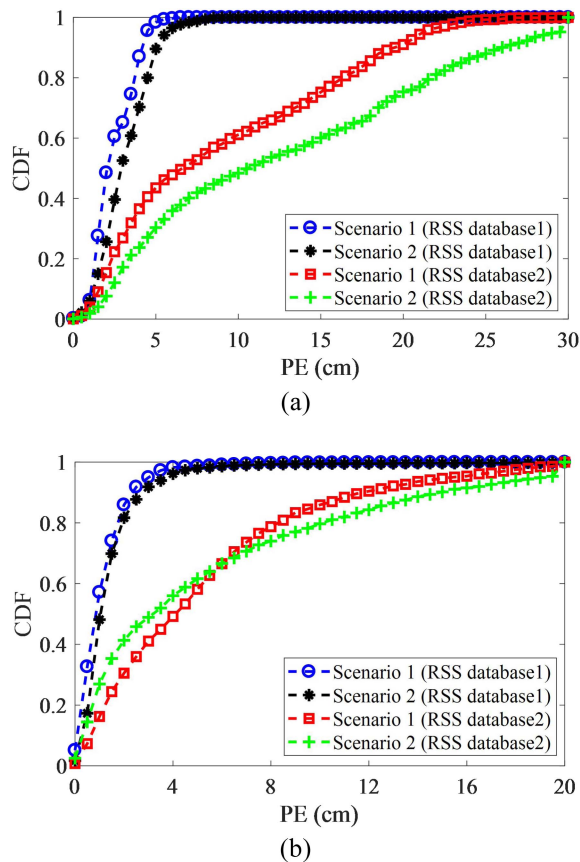


Fig. 9. CDF of PE across different natural light intensities. (a) WKNN. (b) Proposed Algorithm 1.

To verify the effectiveness of the proposed RSS database construction method in mitigating the effects of natural light, we compared its performance with traditional RSS database construction methods that do not account for natural light interference [11], [12], [16], [20]. Additionally, WKNN [7], [8], [11], [13], [29], [30], a commonly used ML algorithm in VLP systems was included as a baseline for positioning. The WKNN algorithm uses Euclidean distance for its metric, with the nearest neighbor value set to 4 in the VLP experiment.

Fig. 9 shows that differences in natural light intensity between offline training and online positioning can lead to substantial PEs in VLP systems. In the figure, RSS database1 labels the proposed RSS database construction method, while RSS database2 corresponds to the traditional RSS database construction method, which does not account for the effects of natural light. The results indicate that despite variations in natural light intensity across experimental scenarios and training fingerprints, both the proposed Algorithm 1 and WKNN algorithm achieve comparable localization accuracy when utilizing the proposed RSS database construction method. This is because the proposed method effectively mitigates natural light interference, unlike traditional approaches. When natural light intensity differs significantly between the training and online positioning phases, traditional RSS database construction methods experience higher PEs. These errors increase further as the discrepancy in natural light intensity grows. In Scenario 1 with the WKNN algorithm, the proposed

and traditional RSS database construction methods yield average PEs (APEs) of 2.68 cm and 9.16 cm, respectively. Similarly, in Scenario 1 with the proposed Algorithm 1, the APEs are 1.36 cm and 5.56 cm, respectively, representing a 75.54% reduction in APE for the proposed RSS database construction method. Moreover, Algorithm 1 consistently outperforms WKNN in positioning accuracy, even using the traditional RSS database construction method. Unlike WKNN, which directly relies on the RSS distance metric and is more vulnerable to natural light interference, the proposed algorithm avoids direct dependence on this metric, enhancing its robustness and reliability in varying environmental conditions.

To further evaluate the positioning performance of the proposed VLP algorithm, we compared it against four baseline algorithms: MLP, $RW\ell_1M$, OMP, and WKNN. The baselines are selected for the following considerations. WKNN and MLP algorithms are widely adopted in VLP systems [11], [13]. OMP has become a benchmark in sparse signal recovery due to its computational efficiency [40]. $RW\ell_1M$ can achieve high-precision sparse signal reconstruction through the iterative weight adjustment mechanism [41]. This paper proposes a VLP system based on a novel sparse adaptive CS algorithm.

For comprehensive performance evaluation, we select OMP and $RW\ell_1M$ as baseline comparison methods, as they represent two distinct approaches within the CS framework. Both OMP and $RW\ell_1M$ have been successfully applied in VLP systems [34], [36], [37]. Unless otherwise specified, the sparsity parameter for both $RW\ell_1M$ and OMP was set to 4, while WKNN employed a nearest neighbor value of 4, as these configurations were found to optimize positioning performance for their respective methods. The MLP model architecture comprises an input layer, a hidden layer, and an output layer. Following extensive evaluation for accuracy, the optimal number of hidden nodes was determined to be 20. In the subsequent experimental analysis, the RSS database utilized by the proposed Algorithm 1 and all four baseline algorithms was constructed using the proposed RSS database construction method.

Fig. 10 shows that the proposed VLP method achieves a markedly lower PE across both experimental scenarios than the other four VLP methods. This outcome stems from several factors: OMP and $RW\ell_1M$ use fixed sparsity values, which reduces their robustness in varying localization scenarios. WKNN's accuracy depends on the nearest neighbor parameter and test point distribution, leading to inconsistent performance. In contrast, the proposed VLP method incorporates a sparsity-adaptive approach, dynamically determining the optimal nearest neighbor value for each test point, thus greatly enhancing positioning robustness. The MLP method relies on input data patterns to predict output. With lower training fingerprint density, MLP's localization accuracy is limited.

Tables I and II compare APEs for five VLP methods in scenarios 1 and 2, respectively. Additionally, P_2 and P_4 denote the probabilities that the PE of a test point is 2 cm or less and 4 cm or less, respectively. Results show that the proposed VLP method achieves substantially higher P_2 and P_4 values than the other four methods, particularly in P_2 . The APE of the proposed VLP method is also notably lower than that of the other four methods.

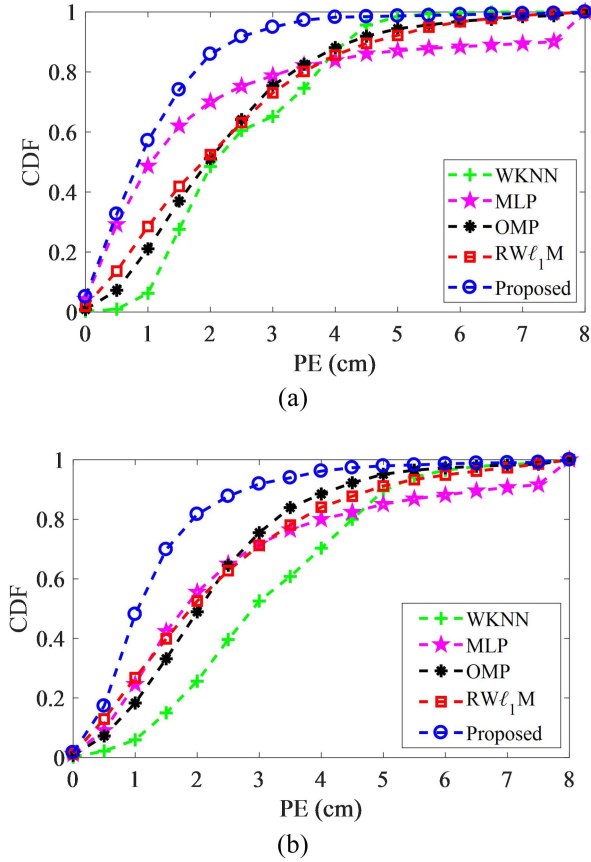


Fig. 10. CDF of PE with a grid spacing at 20 cm. (a) Scenario 1. (b) Scenario 2.

TABLE I
PERFORMANCE COMPARISON IN SCENARIO 1

	APE (cm)	P ₂	P ₄
WKNN	2.68	39.19%	81.38%
MLP	2.77	66.56%	83.25%
OMP	2.49	44.06%	85.94%
RWℓ ₁ M	2.45	47.44%	83.63%
Proposed	1.36	80.94%	97.94%

TABLE II
PERFORMANCE COMPARISON IN SCENARIO 2

	APE (cm)	P ₂	P ₄
WKNN	3.36	19.94%	65.56%
MLP	3.32	49.38%	78.19%
OMP	2.53	40.56%	86.06%
RWℓ ₁ M	2.56	45.75%	80.44%
Proposed	1.70	76.31%	95.06%

For instance, in Table I, the proposed VLP method reduces APE by 49.25%, 50.90%, 45.38%, and 44.49% compared to the WKNN, MLP, OMP, and RWℓ₁M methods, respectively.

The APE of the WKNN, OMP, RWℓ₁M and proposed algorithms varies with the maximum sparsity $K_{\max}$ as shown in Fig. 11. The results indicate that in both experimental scenarios,

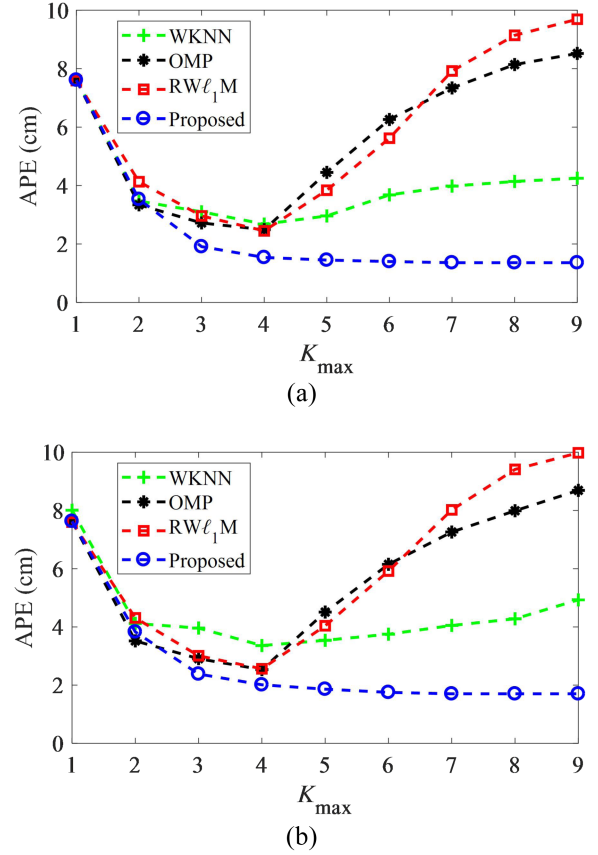


Fig. 11. The impact of $K_{\max}$ on average PEs. (a) Scenario 1. (b) Scenario 2.

the optimal $K_{\max}$ value corresponding to the minimum APE for the OMP, RWℓ₁M, and WKNN algorithms was 4. This is primarily attributed to the ideal square grid structure used in the fingerprint map. However, when the sparsity level $K_{\max}$ exceeds 4 ($K_{\max} > M$), the sparse recovery problem becomes an underdetermined problem, the positioning accuracy of both OMP and RWℓ₁M algorithms degrades as $K_{\max}$ increases further. The main reason is that each OMP iteration involves computing the inner products between all atoms in the dictionary and the current residual, when $K_{\max} > M$, the atoms selected by the OMP algorithm in each iteration may exhibit only weak correlation with the residuals. Once incorrect atoms are included in the support set, the residual update is computed based on an erroneous subspace, leading to error propagation. Similarly, RWℓ₁M optimizes the ℓ_1 -norm through iterative weight adjustments to approximate ℓ_0 sparsity constraints. However, when $K_{\max} > M$, the ℓ_1 -relaxed convex optimization fails to guarantee sparsity, and the reweighting process amplifies noise instead of authentic signal components. These combined effects lead to error accumulation and reconstruction accuracy deterioration. In contrast, the APE of the proposed algorithm decreases gradually with increasing $K_{\max}$ until convergence.

Fig. 12 presents the heatmap of optimal sparsity values K for the proposed Algorithm 1. The results reveal that the optimal sparsity distribution for 1600 test points is predominantly concentrated between 4 and 7 across the two experimental scenarios.

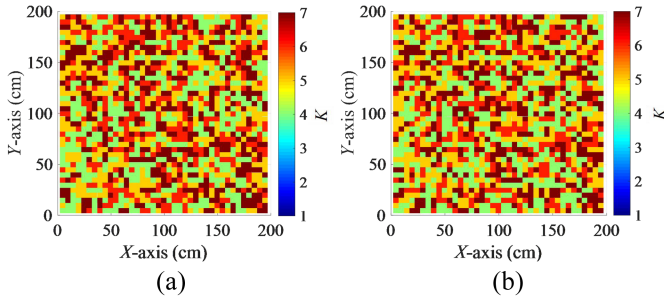


Fig. 12. The heatmap of the optimal sparsity K for each test point. (a) Scenario 1. (b) Scenario 2.

TABLE III
COMPUTATIONAL COMPLEXITIES

Algorithm	Theoretical complexity	Average CPU running time
WKNN	$O(KN)$	0.038 ms
MLP	$O(ML+2L)$	0.005 ms
OMP	$O(KMN + KN + 3N)$	0.139 ms
RW ℓ_1 M	$O(TM^2N^{3/2})$	18.667 ms
Proposed	$O(MN + K_{\max}N + 3N)$	0.137 ms

These findings highlight that the optimal sparsity for each test point is not fixed. Employing a fixed sparsity can compromise the algorithm's robustness, as incorrect selections may result in significant PEs.

Table III summarizes the theoretical computational complexities and average CPU running times for the five VLP methods. The computational complexity of the WKNN algorithm mainly arises from selecting the K smallest values among N RSS distances, yielding a complexity of $O(KN)$. For the MLP algorithm, the computational complexity depends on the neuron counts in the input layer, hidden layer, and output layer, represented by M , L , and 2, respectively, resulting in an $O(ML+2L)$ complexity. The weighted ℓ_1 -norm minimization problem can be solved efficiently by linear programming techniques, and its computational complexity is $O(M^2N^{3/2})$ when using the interior point method [42]. Assuming the iterative convergence requires T iterations, the RW ℓ_1 M algorithm has a complexity of $O(TM^2N^{3/2})$. Among these, the MLP algorithm shows the lowest average CPU running time. However, all five VLP algorithms exhibit average CPU running times that remain within a computationally affordable level. In summary, the proposed VLP algorithm achieves high positioning accuracy with exceptionally low computational cost.

To evaluate how grid spacing S affects positioning performance, the grid spacing of the training fingerprints was increased from 20 cm to 40 cm while keeping the test point distribution in Fig. 6 unchanged. In this case, the number of training fingerprints was reduced from 121 to 36. Positioning results are presented in Fig. 13. The results show that the proposed VLP method achieves significantly lower PE than the other four VLP methods with grid spacing at 40 cm. For instance, In scenario 1, the APEs for WKNN, MLP, OMP, RW ℓ_1 M, and the proposed method are

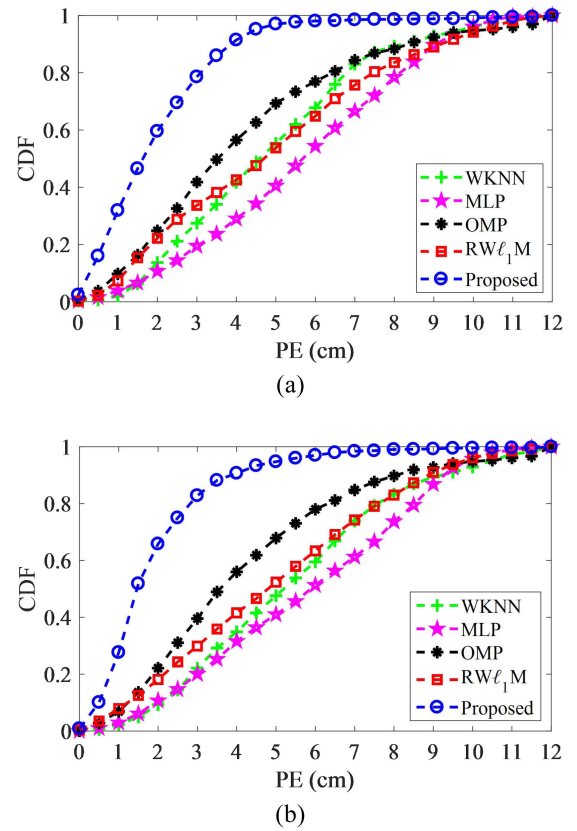


Fig. 13. CDF of PE with a grid spacing at 40 cm. (a) Scenario 1. (b) Scenario 2.

5.07 cm, 5.88 cm, 4.47 cm, 5.09 cm, and 2.15 cm, respectively. In scenario 2, the corresponding APEs are 5.63 cm, 6.02 cm, 4.56 cm, 5.18 cm, and 2.21 cm.

In a practical VLP application scenario, the positions of training fingerprints, as shown in Fig. 6, may not form an ideal, uniformly spaced structure. To evaluate the robustness of the proposed VLP algorithm, we randomly select a subset of 121 training fingerprints with a grid spacing of 20 cm from Fig. 6 for performance evaluation. Let Q represent the proportion of randomly chosen training fingerprints. The number of selected training fingerprints is then calculated as $\text{floor}(NQ)$, where floor denotes rounding down to the nearest integer. To account for randomness, the APE is determined by positioning 1600 test points 100 times with the same Q value. Fig. 14 shows how varying the random selection rate of training fingerprints influences average PEs across five VLP methods. Due to the robustness of the constructed RSS database, the proposed method maintains nearly consistent positioning performance across varying training fingerprint selection rates Q in both scenarios. Among the five VLP methods, the WKNN algorithm is most sensitive to changes in Q because it directly relies on the RSS distance metric. Similarly, the MLP method's accuracy depends on input data density, so low-density fingerprints lower its prediction precision and lead to larger PEs. In contrast, the three CS-based VLP methods (proposed, OMP, and RW ℓ_1 M) are less affected by changes in Q . Thanks to its adaptive sparsity, the proposed method achieves notably lower PEs than both OMP and RW ℓ_1 M.

TABLE IV
PERFORMANCE COMPARISON WITH OTHER VLP SYSTEMS

VLP system	APE	Ambient light	Computational complexity	Storage complexity	Grid spacing
TinyML [3]	1.7 cm	No	$O(TM \log(N))$	$O(T2^d + MN)$	1 cm
Meta-heuristic [12]	1.74 cm	No	$O(KQ_1G_1)$	$O(KQ_1 + MN)$	20 cm
FusionVLP [17]	8.67 cm	Yes	$O(pN_1g^3 + c^3 + N_1gc)$	$O(pN_1g)$	
RatioVLP [19]	8.78 cm	Yes	$O(MN_2(2M)^2)$	$O(MN_2(2M))$	
XGBoost [28]	2.1 cm	No	$O(TMN)$	$O(T2^d + MN)$	10 cm
MPPC [29]	7.48 cm	Yes	$O(T_hKN)$	$O(MN)$	10 cm
CPC [30]	7.17 cm	Yes	$O(G_2MN + KN)$	$O(2MN)$	50 cm
DeepVLP [32]	7.69 cm	Yes	$O(L_1(nv^2 + Ev))$	$O(L_1nv + n^2 + L_1v^2)$	20 cm
Proposed	1.36 cm 2.15 cm	Yes	$O(MN + K_{\max}N + 3N)$	$O(MN)$	20 cm 40 cm

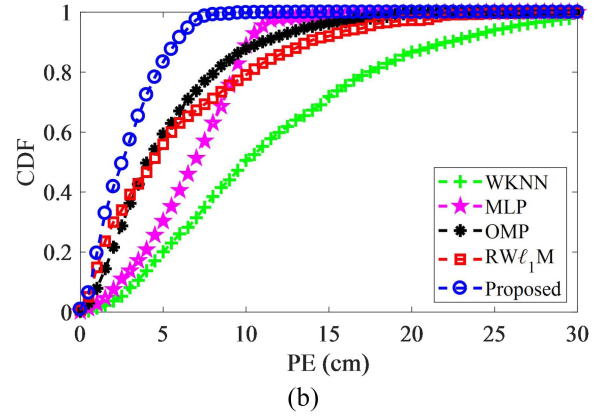
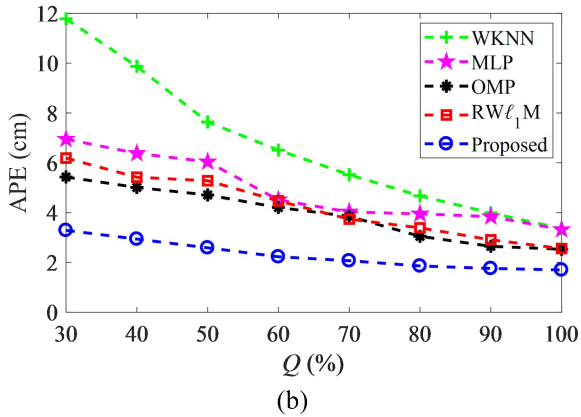
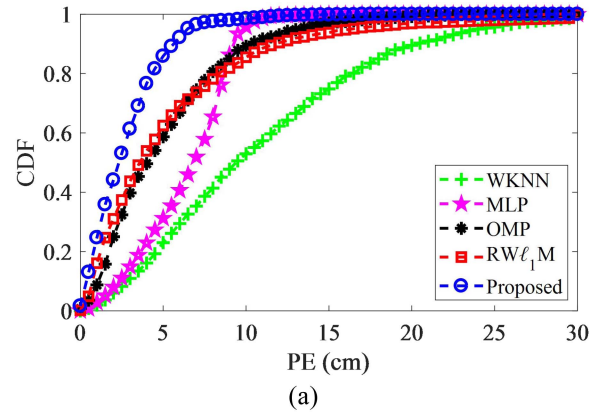
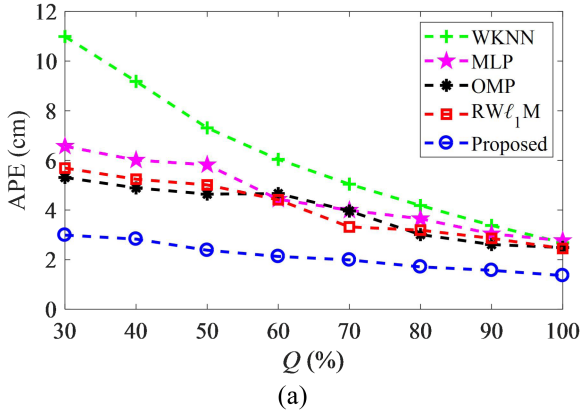


Fig. 14. Effect of the random selection rate Q of training fingerprints on average PEs. (a) Scenario 1. (b) Scenario 2.

Fig. 15 demonstrates that with a Q value of 30%, i.e., the number of training fingerprints is 36, the proposed VLP method achieves significantly lower PEs than the other four VLP methods. In scenario 1, the APEs for WKNN, MLP, OMP, $RW\ell_1M$, and the proposed method are 10.99 cm, 6.57 cm, 5.31 cm, 5.69 cm, and 2.99 cm, respectively. In scenario 2, the corresponding APEs are 11.78 cm, 6.95 cm, 5.43 cm, 6.19 cm, and 3.12 cm.

To evaluate potential positioning accuracy improvements at finer resolutions, we assessed the proposed VLP system's performance across three grid spacings (corresponding to three distinct CS matrix sizes). As shown in Fig. 16, the CDF of PEs shows

Fig. 15. CDF of PE for a training fingerprint random selection rate Q of 30%. (a) Scenario 1. (b) Scenario 2.

marked improvement as grid spacing reduces, corresponding to increased column dimensionality in the CS matrix. This trend occurs primarily because reducing the grid spacing increases the density of training fingerprints. As indicated in (9), this higher fingerprint density allows the system to more accurately identify the K most relevant training fingerprints, thereby effectively minimizing PEs. It should be noted that although increasing the dimensionality of the CS matrix enhances positioning accuracy, this improvement comes at the expense of greater computational burden and storage capacity requirements.

To validate the applicability of the proposed method in 3D positioning, we extended the 2D fingerprint positioning approach

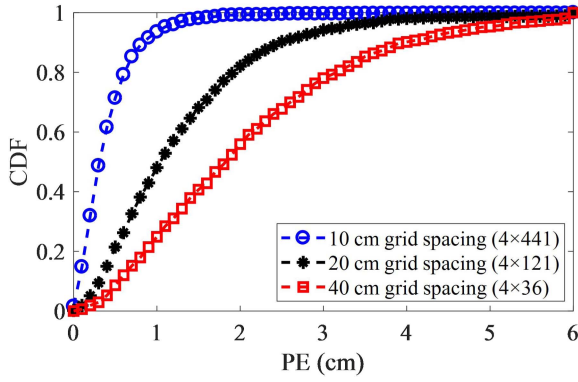


Fig. 16. CDF of PEs versus grid spacing (corresponding CS matrix sizes) in the proposed VLP system.

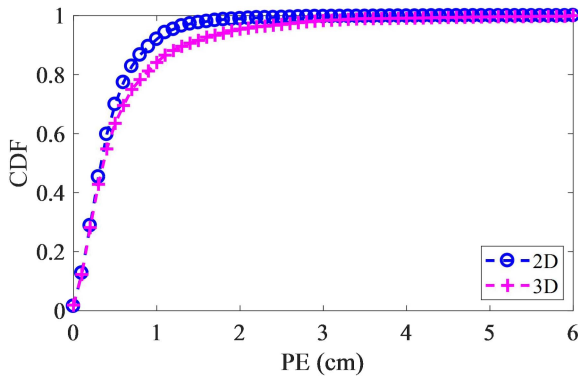


Fig. 17. CDF of PEs comparing 2D and 3D performance in the proposed VLP system.

to a 3D space. The training fingerprints were arranged in a 3D cubic grid with 10 cm spacing along all three axes (X , Y , and Z). A total of 882 training points were distributed across the 3D space. Similarly, the test points followed the same 3D grid structure, i.e., 3200 test points in total, for a consistent assessment. As shown in Fig. 17, the CDF of 3D PEs is marginally lower than that of 2D PEs, primarily due to the introduction of Z -axis errors. Nevertheless, both methods maintain strong consistency in their positioning results overall. This demonstrates the effective extensibility of the proposed approach to 3D positioning scenarios. The fundamental rationale stems from the method's core principle of selecting the K most relevant RSS fingerprints for any test point. The extension from 2D to 3D space remains valid because, with an appropriately configured LED array, each 3D coordinate maintains a unique RSS vector signature – analogous to the 2D scenario. By (9), our system identifies the K most relevant training fingerprints, guaranteeing accurate positioning in both 2D and 3D environments.

For assessing how LED layouts affect the positioning performance of the proposed VLP system, with the grid spacing set at 10 cm, we tested multiple arrangements: (i) linear (2 LEDs), (ii) triangular (3 LEDs), and (iii) quadrilateral patterns (square and rhombus with 4 LEDs). As shown in Fig. 18, when using a linear layout, geometric constraints make the RSS vectors difficult to distinguish along the direction perpendicular to the LED connection line. This leads to a loss of spatial uniqueness

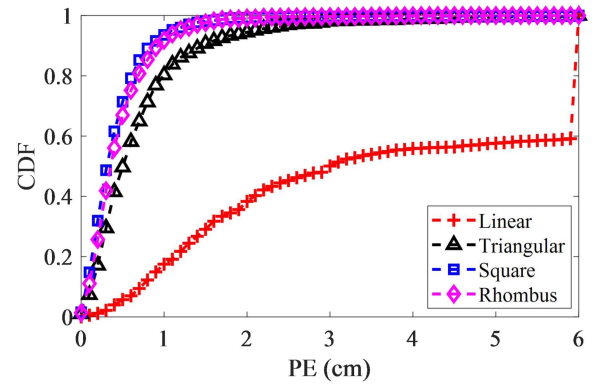


Fig. 18. CDF of PEs for different LED layouts in the proposed VLP system.

in RSS fingerprints, ultimately causing significant PEs. When using a triangular layout, the RSS vector at each position exhibits a unique RSS vector signature in the 2D space. This configuration provides sufficient spatial resolution capability, enabling high positioning accuracy. When the number of LEDs increases to four (including square and rhombus layouts), the CDF of PEs shows no significant difference. This suggests that while maintaining uniform illumination, increasing the number of light sources can effectively reduce PEs.

Table IV compares the performance of various experimentally validated VLP systems. Since positioning accuracy depends on multiple factors (e.g., ranging model, grid spacing, ambient light, receiver type, and algorithm), the APE values in Table IV are sourced from original studies. In Table IV, T and d are the number of trees and the depth of each tree in RF and XGBoost models, respectively. In meta-heuristic, Q_1 represents the population size, G_1 represents the number of iterations. For FusionVLP, N_1 , g , c and p correspond to the ensemble size, state dimension, observation dimension, and lag window size. N_2 is the number of calibration points. T_h is the number of hyperparameter combinations. G_2 is the number of optimization steps. For DeepVLP, n , E , v , and L_1 correspond to the node count, edge count, feature dimension, and layer count. Ranging-based methods like FusionVLP, RatioVLP, and DeepVLP eliminate the need for a full fingerprint database. Nevertheless, model parameter calibration is still needed, which adds some computational and storage complexity. In the TinyML approach, maintaining a grid spacing of 1 cm leads to significantly higher computational and storage costs. Due to the 10 cm grid spacing, both the XGBoost and MPPC methods exhibit high computational and storage costs. The meta-heuristic-based method determines optimal nearest neighbor weights through population cooperative optimization. Consequently, its computational complexity is extremely high. In practical applications, the computational and storage complexity of the proposed VLP system mainly depends on the number of training fingerprints (N). Even under the influence of natural light, it achieves an APE of 2.15 cm at a 40 cm grid spacing. Note that while reducing the grid spacing (i.e., increasing N) can improve positioning accuracy in fingerprint-based VLP systems, this improvement inevitably increases both computational

complexity and storage complexity. Despite this trade-off, the proposed VLP system achieves excellent positioning accuracy while maintaining low computational and storage complexity.

IV. CONCLUSION

To enhance the robustness of VLP systems under natural lighting conditions, this paper presents a novel RSS-based database construction method. The approach leverages PWM to reduce natural light interference and introduces a low-complexity, sparsity-adaptive CS technique. The proposed method enables accurate signal reconstruction without requiring prior knowledge of their sparsity. Experimental results confirm that the proposed method effectively mitigates the impact of natural light. With a grid spacing of 20 cm, the proposed method achieves an APE of 1.36 cm, representing a 75.54% reduction compared to the traditional RSS database construction method. Furthermore, the proposed VLP system maintains an APE of 2.15 cm even at a 40 cm grid spacing. This demonstrates that our system achieves excellent positioning accuracy while maintaining low computational and storage complexity.

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